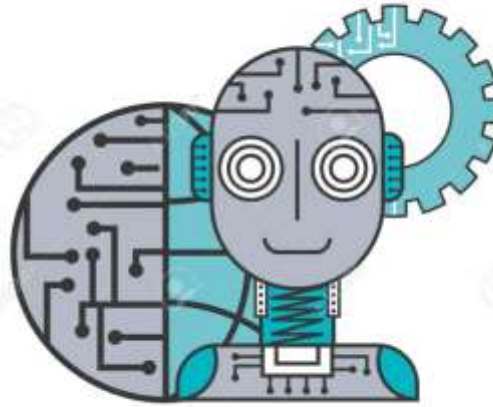




Clustering Problems

Ioannis Tsioulis (MC)

Declaration

This slide is created based on

Books:

- *Data Mining Concept and Technologies, Chapter 10.*
- *Doing Data Science, Chapter 3.*

Online sources:

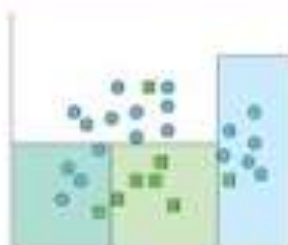
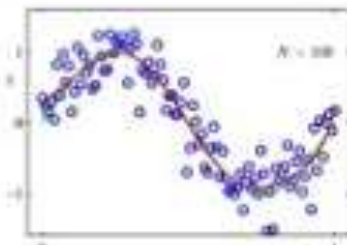
- ‘Unsupervised Machine Learning: Clustering Analysis’, Victor Roman, Mar, 2019
<https://towardsdatascience.com/unsupervised-machine-learning-clustering-analysis-d40f2b34ae7e>
- ‘Unsupervised Learning and Data Clustering’ Sanatan Mishra, May, 2017
<https://towardsdatascience.com/unsupervised-learning-and-data-clustering-eeecb78b422a>

Agenda

- Clustering Analysis
- Unsupervised Learning
- K-mean algorithm
- Advanced clustering algorithm

What is Clustering Analysis?

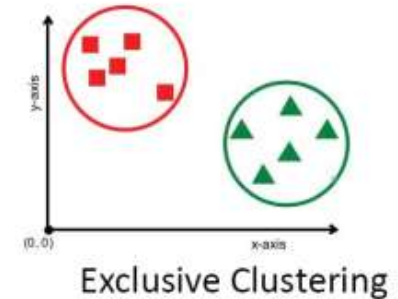
Definition: Clustering analysis or simply clustering is the process of partitioning a set of data objects into subset. Each subset is a cluster, such that objects in a cluster are similar to one another, yet dissimilar to objects in other clusters. (The data objects are **un-labelled** and **requiring Unsupervised Learning**)

Predictive methods	Descriptive methods
Classification  Learns a method for predicting the instance class from pre-labeled (classified) instances	Clustering  Finds "natural" grouping of instances given un-labeled data
Regression  An attempt to predict a continuous attribute	Association Rules  Method for discovering interesting relations between variables in large DBs

Classify Clustering Problems

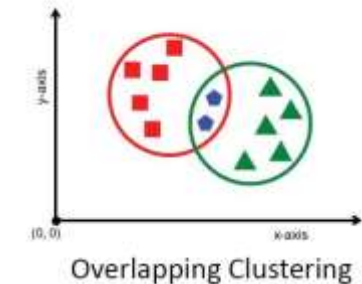
- Exclusive Clustering:

It is a type of hard Clustering in which the data points or items are exclusive to one cluster.



- Overlapping Clustering:

It is a type of Soft Clustering in which the data point or item belongs to multiple clusters.

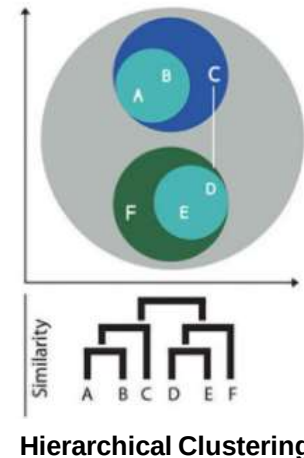


- Hierarchical Clustering:

Each observation is taken as a separate cluster, and all the similar clusters are grouped together in a hierarchical manner resulting in a complete cluster.

- Probabilistic Clustering

the assignment of points to clusters is “soft”, in the sense that the membership of a data point x in a **cluster** C_k is given as a probability, denoted by $p_k(x)$.



Agenda

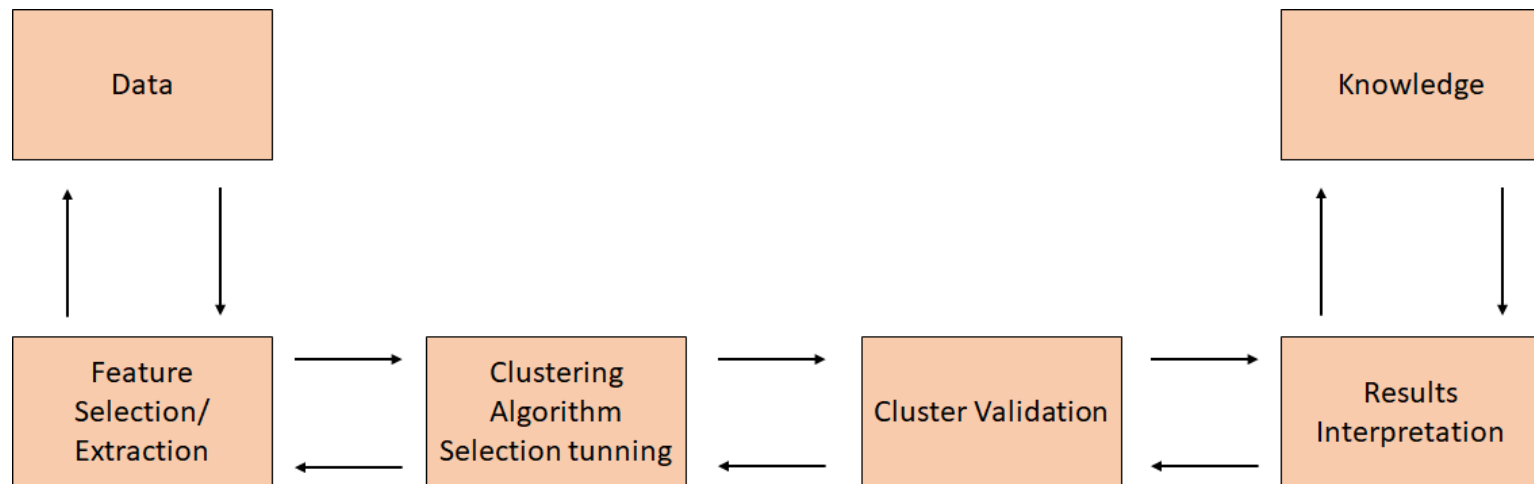
- Clustering Analysis
- Unsupervised Learning
- K-mean algorithm
- Advanced clustering algorithm

Introduction to Unsupervised Learning

- So far we've only seen supervised learning, where we know beforehand what label (aka the "right answer") is and we're trying to get our model to be as accurate as possible, defined by our chosen evaluation metric.
- However, when dealing with real-world problems, most of the time, data will **not come with predefined labels**, so we will want to develop machine learning models that can classify correctly this data, by finding by **themselves some commonality in the features**, that will be used to predict the classes on new data.

Unsupervised Learning Analysis Process

The overall process that we will follow when developing an unsupervised learning model can be summarized in the following chart:



Picture Source: 'Unsupervised Machine Learning: Clustering Analysis', Victor Roman, Mar, 2019
<https://towardsdatascience.com/unsupervised-machine-learning-clustering-analysis-d40f2b34ae7e>

Issues and applications with Unsupervised Learning

Issues:

- Unsupervised Learning is harder as compared to Supervised Learning tasks..
- How do we know if results are meaningful since no answer labels are available?
- Let the expert look at the results (external evaluation)
- Define an objective function on clustering (internal evaluation)

Unsupervised learning main applications are:

- Segmenting datasets by some shared attributes.
- Detecting anomalies that do not fit to any group.
- Simplify datasets by aggregating variables with similar attributes.

Why Unsupervised Learning is needed

- Annotating large datasets is very costly and hence we can label only a few examples manually. Example: Speech Recognition
- There may be cases where we don't know how many/what classes is the data divided into. Example: Data Mining
- We may want to use clustering to gain some insight into the structure of the data before designing a classifier or deal with missing data.

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Common Clustering Algorithms

- K-means
- Fuzzy K-means
- Hierarchical clustering
- DBSCAN
- Mixture of Gaussians(

<https://www.youtube.com/watch?v=DODphRRL79c>)

K-mean Algorithm Example

- k-means is the first *unsupervised* learning technique we'll look into, where the goal of the algorithm is to determine the definition of the right answer by finding clusters of data for you.
- Let's say you have some kind of data at the user level, e.g., Google+ data, survey data, medical data, or SAT scores.
- Start by adding structure to your data. Namely, assume each row of your dataset corresponds to a user as follows:

age gender income state household size

Your goal is to *segment* the users. This process is known by various names: besides being called segmenting, you could say that you're going to *stratify*, *group*, or *cluster* the data. They all mean finding similar types of users and bunching them together.

Example Motivation

Why would you want to do this? Here are a few examples:

- You might want to give different users different experiences. Marketing often does this; for example, to offer toner to people who are known to own printers.
- You might have a model that works better for specific groups. Or you might have different models for different groups.
- Hierarchical modelling in statistics does something like this; for example, to separately model geographical effects from household effects in survey results.

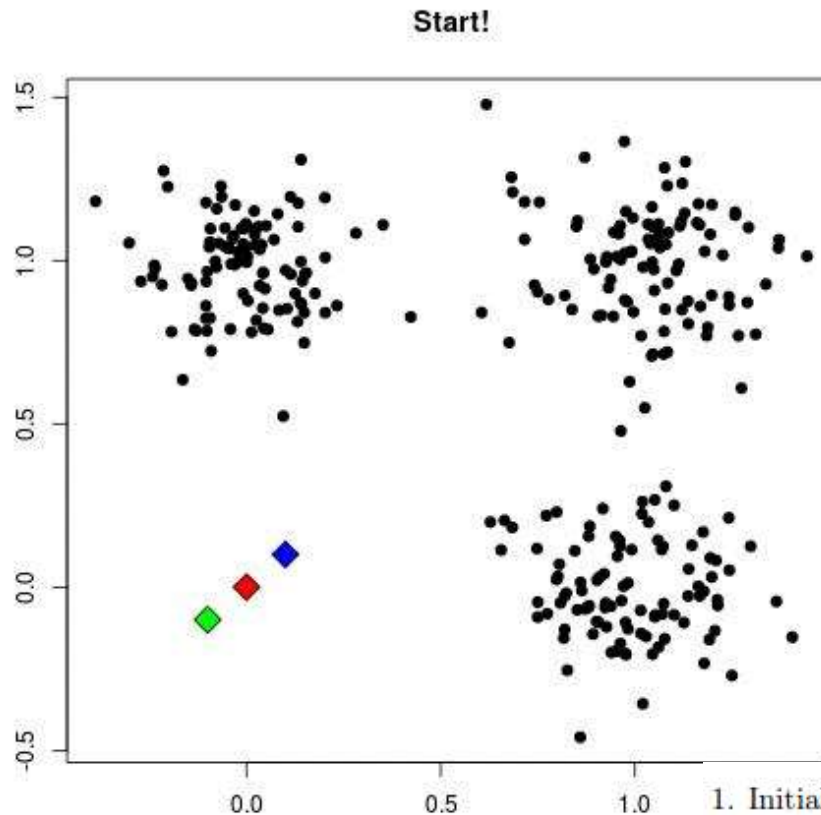
EDA and Data-preprocessing When Applying K-means

- Features must be measured on the same scale, so it may be necessary to perform z-score standardization or max-min scaling.
- When dealing with categorical data, we will use the get dummies function.
- Exploratory Data Analysis (EDA) is very helpful to have an overview of the data and determine if K-Means is the most appropriate algorithm.
- The mini-batch method is very useful when there is a large number of columns, however, it is less accurate.

K-Means Clustering Explanation

- The main idea is to define k centres, one for each cluster. These centroids should be placed in a smart way because of different location causes different result. So, the better choice is to place them as much as possible far away from each other.
- The next step is to take each point belonging to a given data set and associate it to the nearest centroid.
- When no point is pending, the first step is completed and an early groupage is done.
- At this point we need to re-calculate k new centroids as barycentre of the clusters resulting from the previous step.
- After we have these k new centroids, a new binding has to be done between the same data set points and the nearest new centroid. A loop has been generated.
- As a result of this loop we may notice that the k centroids change their location step by step until no more changes are done. In other words centroids do not move any more.

K-Means in Motion



1. You define how many cluster you would like to have (k) and the algorithm will randomly select the position of k new centroids.
2. The next step is to take each point belonging to a given data set and associate it to the nearest centroid.
3. When no point is pending, the first step is completed and an early groupage is done. we need to re-calculate k new centroids as barycentre of the clusters resulting from the previous step.
4. After we have these k new centroids, a new binding has to be done between the same data set points and the nearest new centroid. A loop has been generated.

1. Initialize **cluster centroids** $\mu_1, \mu_2, \dots, \mu_k \in \mathbb{R}^n$ randomly.

2. Repeat until convergence: {

For every i , set

$$c^{(i)} := \arg \min_j \|x^{(i)} - \mu_j\|^2.$$

For each j , set

$$\mu_j := \frac{\sum_{i=1}^m 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^m 1\{c^{(i)} = j\}}.$$

}

K-Means Hyper-parameters

- Number of clusters: The number of clusters and centroids to generate.
- Maximum iterations: Of the algorithm for a single run.
- Number initial: The number of times the algorithm will be run with different centroid seeds. The final result will be the best output of the number defined of consecutive runs, in terms of inertia.

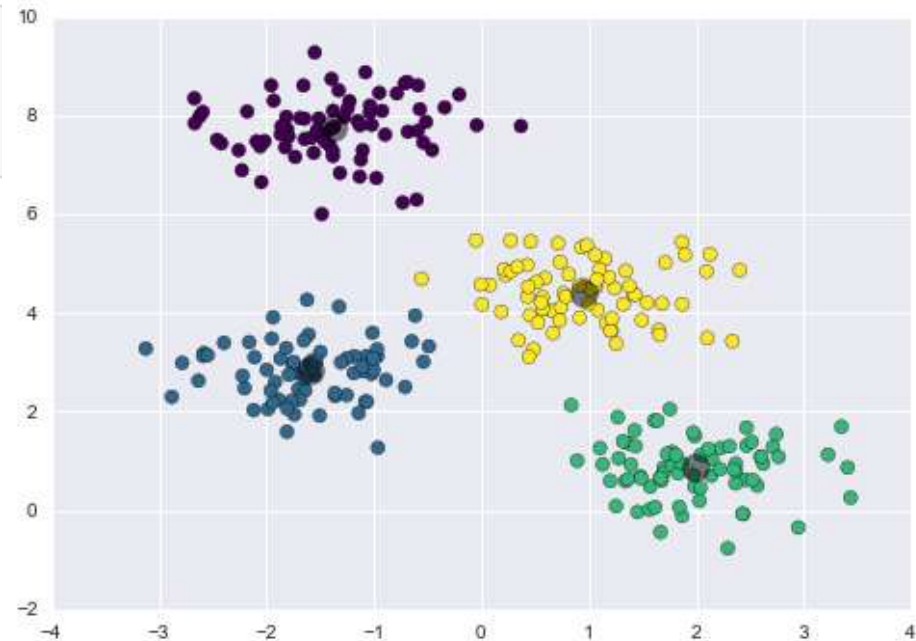
K-mean Implementation in Python

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

```
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns; sns.set() # for plot styling
import numpy as np
```

```
from sklearn.datasets.samples_generator import make_blobs
X, y_true = make_blobs(n_samples=300, centers=4,
                        cluster_std=0.60, random_state=0)
plt.scatter(X[:, 0], X[:, 1], s=50);
```

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=4)
kmeans.fit(X)
y_kmeans = kmeans.predict(X)
```



<https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

Challenges of K-Means

- **How to Choose the Right K Number**
- The output for any fixed training set won't be always the same, because the initial centroids are set randomly and that will influence the whole algorithm process.
- As stated before, due to the nature of Euclidean distance, it is not a suitable algorithm when dealing with clusters that adopt non-spherical shapes.

How to Choose the Right K Number

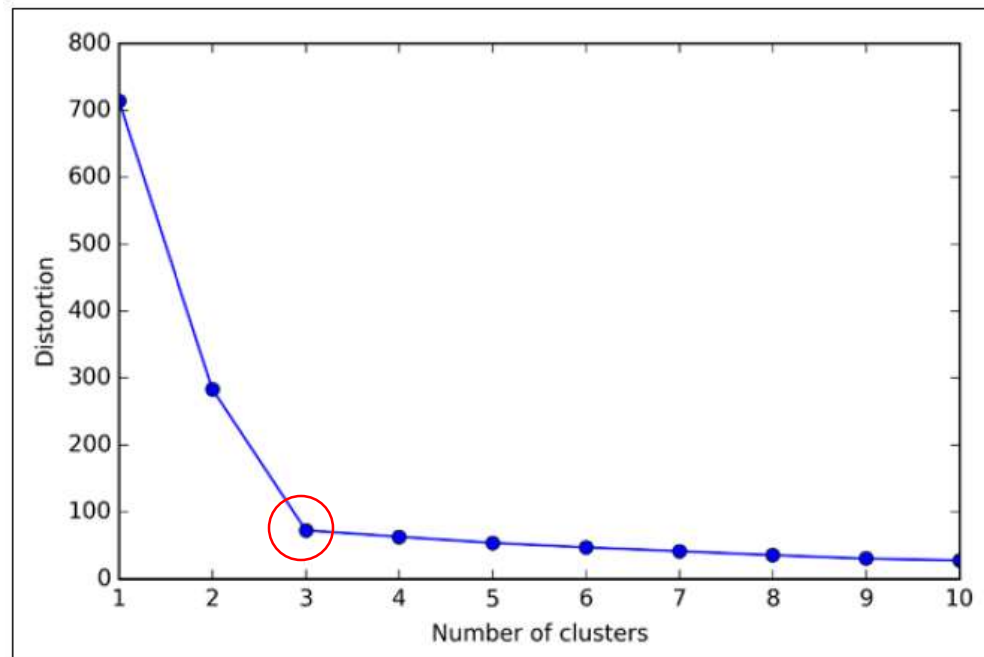
- Choosing the right number of clusters is one of the key points of the K-Means algorithm. To find this number there are some methods:
- Field knowledge
- Business decision
- Elbow Method

Elbow Method for K selection

The elbow method is used for determining the correct number of clusters in a dataset. It works by plotting the ascending values of K versus the total error obtained when using that K.

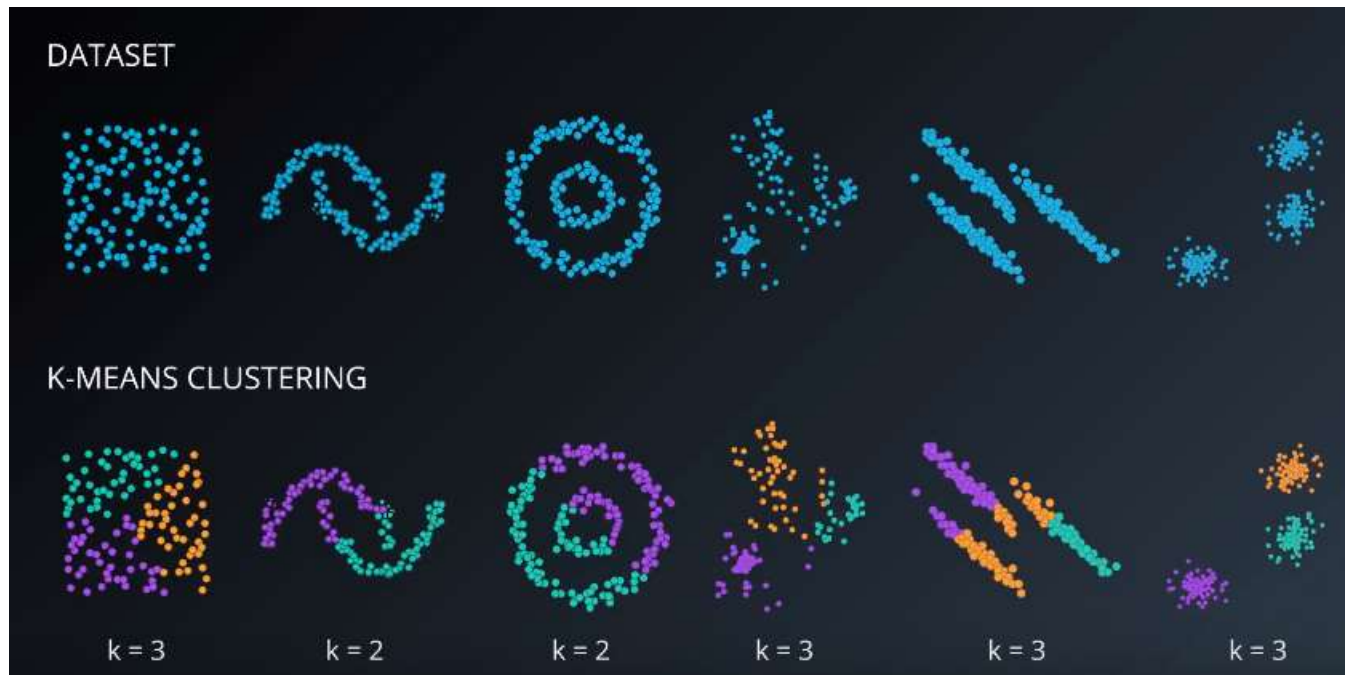
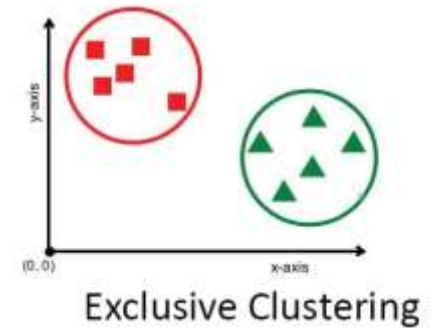
$$\% \text{ Variance} = \frac{\text{Variance between groups}}{\text{Total variance}}$$

The goal is to find the k that for each cluster will not rise significantly the variance also consider trade off balance of computing cost and accuracy



K-Means Limitations

- Although K-Means is a great clustering algorithm, it is most useful to deal with exclusive clustering problems.
- The following picture show what we would obtain if we use K-means clustering in each dataset even if we knew the exact number of clusters beforehand:



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Advanced Clustering Algorithms

- Fuzzy K-Means Clustering
- Hierarchical Clustering
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

Fuzzy K-Means Clustering

The fuzzy k-means algorithm is the following:

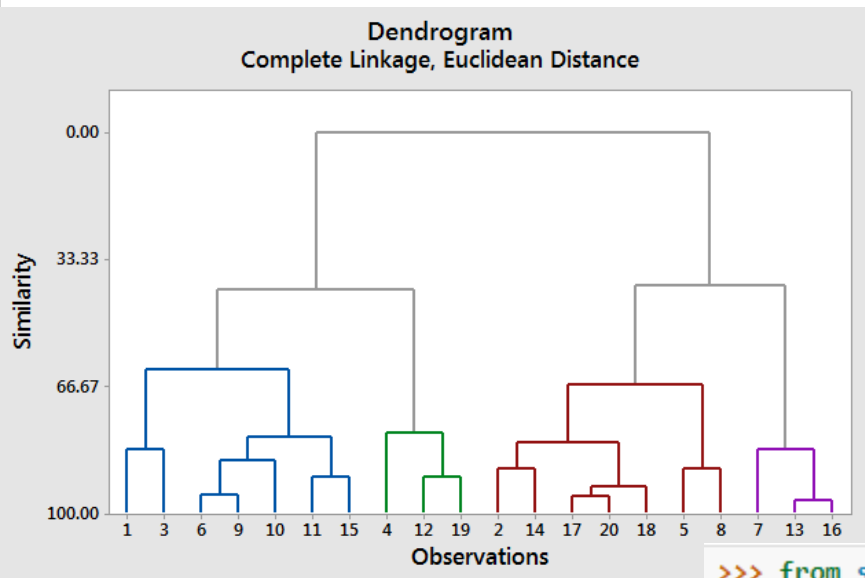
- **Assume** a fixed number of clusters K .
- **Initialization:** Randomly initialize the k-means μ_k associated with the clusters and compute the probability that each data point X_i is a member of a given cluster K ,
 $P(\text{Point } X_i \text{ Has Label } K | X_i, K)$.
- **Iteration:** Recalculate the centroid of the cluster as the weighted centroid given the probabilities of membership of all data points X_i :

$$\mu_k(n+1) = \frac{\sum_{x_i \in k} x_i \times P(\mu_k | x_i)^b}{\sum_{x_i \in k} P(\mu_k | x_i)^b}$$

- **Termination:** Iterate until convergence or until a user-specified number of iterations has been reached (the iteration may be trapped at some local maxima or minima)

Hierarchical Clustering

- Hierarchical clustering is an alternative to prototype-based clustering algorithms. The main advantage of Hierarchical clustering is that we do not need to specify the number of clusters, it will find it by itself. In addition, it enables the plotting of dendrograms. Dendrograms are visualizations of a binary hierarchical clustering.



```
>>> from sklearn.cluster import AgglomerativeClustering
>>> import numpy as np
>>> X = np.array([[1, 2], [1, 4], [1, 0],
...               [4, 2], [4, 4], [4, 0]])
>>> clustering = AgglomerativeClustering().fit(X)
>>> clustering
AgglomerativeClustering()
>>> clustering.labels_
array([1, 1, 1, 0, 0, 0])
```

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>

Cons and Pros of Hierarchical Clustering

Advantages of Hierarchical Clustering

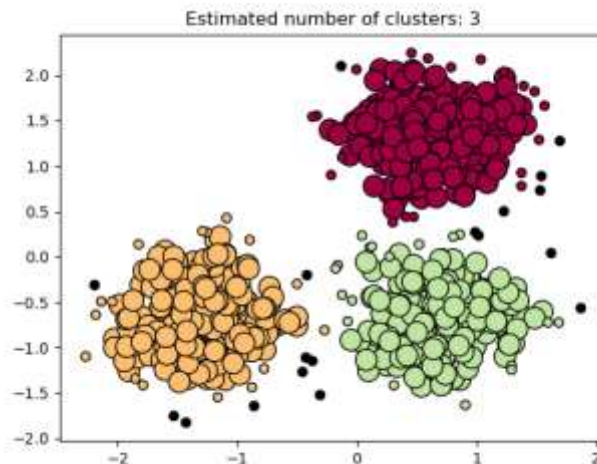
- The resulting hierarchical representations can be very informative.
- Dendograms provide an interesting and informative way of visualization.
- They are specially powerful when the dataset contains real hierarchical relationships.

Disadvantages of Hierarchical Clustering

- They are very sensitive to outliers and, in their presence, the model performance decreases significantly.
- They are very expensive, computationally speaking.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

Density-Based Spatial Clustering of Applications with Noise, or DBSCAN, is another clustering algorithm specially useful to correctly identify noise in data. It is based on a number of points with a specified radius ϵ and there is a special label assigned to each data point.



```
from sklearn.cluster import DBSCAN  
db = DBSCAN(eps=0.3, min_samples=10).fit(X)
```

Cons and Pros of DBSCAN

DBSCAN Advantages

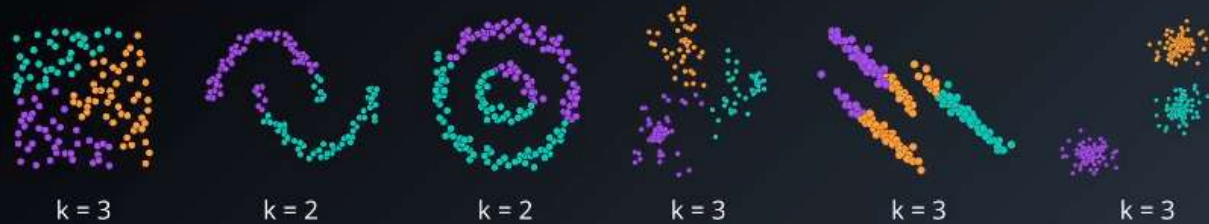
- We do not need to specify the number of clusters.
- There is high flexibility in the shapes and sizes that the clusters may adopt.
- It is very useful to identify and deal with noise data and outliers.

DBSCAN Disadvantages

- It faces difficulties when dealing with boarder points that are reachable by two clusters.
- It doesn't find well clusters of varying densities.

Comparing

K-MEANS CLUSTERING



SINGLE LINK HIERARCHICAL CLUSTERING



DBSCAN



Summary

In this lecture, you should understand:

- The concepts of clustering analysis and unsupervised learning
- The algorithms of unsupervised learning, especially the most basic one – K-mean.
- The practice session, you will see a demonstration of a K-mean algorithm implementation and a use case for you to provide a K-mean solution.

Next:

- Regression problem and algorithms
- Practices on regression algorithm implementation

Further recommended reading

(Machine) Learning by Example: Clustering

<https://medium.com/opex-analytics/machine-learning-by-example-clustering-a4ff01d81d3a>

Book

*Data Mining Concept and Technologies,
Chapter 10.*

Doing Data Science, Chapter 3.